

Auditing System-1 Models on Biosecurity-Relevant Benchmarks: Calibration, Selective Prediction, and Permutation Instability in a Non-Generative Model

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Abstract—Non-generative “System-1” models return structured probabilistic decisions in a single forward pass, without autoregressive decoding, at a small fraction of the inference cost of a generative model. This makes them of interest as inexpensive components in larger pipelines, but their reliability on biosecurity-relevant tasks has not been systematically examined. We audit one commercial System-1 model on 6,020 multiple-choice items drawn from the Weapons of Mass Destruction Proxy (WMDP), a paraphrase-robust WMDP-Bio variant, and six LAB-Bench subtasks, measuring accuracy, calibration, error detection, selective prediction, and sensitivity to the order in which answer options are presented. Accuracy is strongly task-dependent. Once the vendor’s uncertainty field is correctly interpreted, the model is reasonably well calibrated (pooled expected calibration error 0.034) and its top-1 probability separates correct from incorrect predictions (pooled AUROC 0.820), though both degrade substantially on the weaker tasks. Under four cyclic rotations of the answer options, 37.4% of WMDP-Cyber items receive different answers; a control using byte-identical repeated calls attributes most of this to option order rather than run-to-run variation. Averaging probabilities across rotations improves WMDP-Cyber accuracy by +3.8 percentage points, and applying it only to low-confidence items recovers most of that gain at well under the cost of averaging every item.

Index Terms: model evaluation, calibration, selective prediction, robustness, biosecurity, AI safety

1. Introduction

Recent work on efficient inference has produced models that return a typed, probabilistic decision from a single forward pass rather than by decoding tokens autoregressively. Such “System-1” models are offered for settings in which many decisions must be made cheaply. The evaluation reported here cost 0.57 USD for 19,060 calls. Answering the same items with a generative model at current list prices would cost between roughly 0.71 USD, using the

cheapest small model and emitting only the option label, and 421 USD at frontier prices with a short chain of thought (Section 3). That difference is what makes repeated evaluation of the same item a practical experimental option here rather than a hypothetical one.

Low-cost probabilistic models are potentially relevant to AI safety and biosecurity pipelines, where auxiliary classifiers are already deployed alongside larger models [1], [2]. Before such a role could be considered, the basic empirical properties of these models need to be established: how accurate they are, how well their reported uncertainty tracks correctness, whether that uncertainty supports abstention, and how stable their decisions are under changes that do not alter the meaning of the input. Benchmark accuracy addresses only the first of these.

This paper reports a reliability audit of one commercially available System-1 model, JEV, on two kinds of benchmark. The Weapons of Mass Destruction Proxy (WMDP) [3] is a multiple-choice set covering biosecurity, chemical security, and cybersecurity, written as a measurable stand-in for knowledge that could assist misuse and introduced so that unlearning methods could be scored on *reducing* it; high accuracy on it is therefore not a capability result. LAB-Bench [4] covers practical biology research skill and carries the opposite sign. Neither benchmark asks the model to judge whether a request is dangerous, so neither measures screening or hazard-detection performance. They are used here as subject matter on which reliability, uncertainty, and robustness can be studied.

We ask four questions. When is the model correct? Does its reported uncertainty identify likely errors? Can uncertain items be set aside to raise accuracy on the remainder? And are its decisions stable under a change that does not affect meaning, namely the order in which the answer options are listed?

This paper makes three contributions.

- 1) A reliability audit of a commercial non-generative System-1 model across 6,020 biosecurity- and biology-relevant benchmark items, covering accuracy, calibration, error detection, and selective pre-

diction.

- 2) A controlled answer-order study, with a byte-identical-repeat control that separates option-order sensitivity from run-to-run variation, showing substantial item-level instability despite comparatively stable aggregate accuracy.
- 3) An evaluation of selective permutation averaging, showing that repeated inference applied only to low-confidence items improves accuracy on the instability-heavy task while avoiding the cost of evaluating every item four times.

We do not claim novelty for confidence-based selection of items for further computation [5]–[8], nor for the observation that multiple-choice evaluation is sensitive to option order [9], [10].

2. Background and Related Work

Decision cascades and early exit. Spending extra computation only on inputs a cheap model finds difficult is established: CALM [5] exits a decoder early under a calibrated criterion, Big Little Decoder [6] escalates when a small model’s maximum predicted probability falls below a threshold, and FrugalGPT [7] and RouteLLM [8] allocate queries across models under a cost budget. Our final experiment applies the same idea with additional answer permutations of the same model rather than a larger model. For uncertainty we use standard tools: maximum class probability as an error-detection baseline [11], expected calibration error and reliability diagrams [12], and risk–coverage analysis from selective classification [13].

Multiple-choice robustness. Reordering answer options changes model rankings on leaderboards [9] and degrades accuracy in model-dependent ways [10]; Zheng et al. [14] attribute the effect to a token-level bias toward particular option labels, and formatting changes that preserve meaning can also shift accuracy substantially [15]. That work concerns generative models scored by likelihood over option tokens. We test whether a non-generative model emitting a typed distribution in one pass shows comparable sensitivity.

Safety classifiers and biosecurity context. Deployed language-model systems commonly place auxiliary classifiers around a primary model to detect disallowed content: moderation classifiers [16], input–output safeguards such as Llama Guard [1] and ShieldGemma [17], constitutional classifiers trained to resist jailbreaks [2], with biological risk a specific concern [18]. That work motivates why inexpensive probabilistic models are of interest, but we evaluate none of these safeguard tasks: prior work asks whether a system can block a dangerous request, whereas we ask what reliability properties a cheap System-1 model exhibits on biosecurity-relevant knowledge benchmarks.

3. Experimental Setup

JEV is a commercial non-generative model queried through the `v1` REST endpoint. A request carries a free-

text state and a typed question; the response gives a selected category, a probability distribution over categories, a scalar confidence, the resolved model version, and token usage. All calls used the alias `jev-latest`, which resolved to `jev-1.13.0` throughout, and were made from 2026-09-19 to 2026-09-20. The model is proprietary: no logits, token probabilities, or random seed are exposed, and returned probabilities are quantised to two decimals (101 distinct values observed). Because inference is a single pass rather than autoregressive decoding, per-item cost is low; we did not measure latency and make no claim about it.

Table 1 lists the 10 datasets. WMDP-Bio, WMDP-Chem, and WMDP-Cyber are hazardous-knowledge proxies [3], WMDP-Bio-Robust is a paraphrased variant of WMDP-Bio, and six LAB-Bench subtasks cover practical biology research [4]. Option counts range from 2 to 10, so we report accuracy against each item’s own chance level $1/n$ rather than a single 25% line. LAB-Bench items are formed by combining the reference answer with its distractors under a fixed shuffle seed; items with fewer than two distinct options after de-duplication are skipped. Of 6,021 single-pass calls, 1 returned an API error and is excluded, leaving 6,020 items.

We report accuracy with Wilson intervals; expected calibration error (ECE) over ten equal-width bins; the area under the receiver operating characteristic curve (AUROC) for separating correct from incorrect predictions using $p_{\max}$, the largest returned probability; the error rate conditional on high confidence, $P(\text{error} \mid p_{\max} \geq 0.9)$; and risk–coverage curves obtained by withholding the least confident items. Intervals on proportions are Wilson; intervals on ECE, AUROC, and on differences between accuracies are percentile bootstrap over items, 2,000 resamples with a fixed seed.

For the two four-option WMDP suites we additionally present each item under all four cyclic rotations of its option list (1,273 and 1,987 items, 13,040 calls), mapping each returned distribution back to the original option indices so that rotations are comparable.

The endpoint reports token usage per response. The study consumed 13.5 million input tokens over 19,060 calls, a mean of 709 per call; at 0.042 USD per million input tokens with no output charge, 0.57 USD. The same calls with a generative model, at list prices retrieved 2026-09-20, would cost about 0.71 USD for the cheapest small model emitting only an option label ($1.3\times$), 84 USD mid-tier with a short chain of thought ($148\times$), and 421 USD at frontier prices ($742\times$). These are upper bounds: batch interfaces list at half rate, cached input at a tenth, and large deployments negotiate below list, while an agentic configuration issuing several turns per decision moves the other way. We did not measure latency.

4. Results

4.1. Benchmark performance and uncertainty

Accuracy varies widely across tasks (Table 1). It is highest on WMDP-Bio (85.2%), falls on the paraphrased

TABLE 1. PER-DATASET RESULTS. *Lift* IS CHANCE-ADJUSTED ACCURACY, $(acc - c)/(1 - c)$, USING EACH ITEM’S OWN OPTION COUNT $c = 1/n$ RATHER THAN A SINGLE 25% LINE, SINCE OPTION COUNTS RANGE FROM 2 TO 10. ECE AND AUROC ARE COMPUTED ON $p_{\max}$ RATHER THAN ON THE VENDOR CONFIDENCE FIELD (SECTION 4.1). $P(\text{err} \mid p_{\max} \geq 0.9)$ IS THE ERROR RATE GIVEN A CONFIDENT ANSWER, WITH WILSON INTERVALS. *Retained* IS ACCURACY ON THE ITEMS KEPT WHEN THE LEAST CONFIDENT 40% ARE WITHHELD (SECTION 4.2).

Suite	Dataset	N	opts.	accuracy [95% CI]	lift	ECE	AUROC	$P(\text{err} \mid p_{\max} \geq 0.9)$	retained
WMDP	WMDP-Bio	1273	4	0.852 [0.83, 0.87]	0.802	0.036	0.855	0.039 [0.03, 0.05]	0.971
	WMDP-Bio-Robust	811	4	0.801 [0.77, 0.83]	0.735	0.044	0.823	0.063 [0.04, 0.09]	0.938
	WMDP-Chem	408	4	0.730 [0.69, 0.77]	0.641	0.057	0.824	0.056 [0.03, 0.10]	0.898
	WMDP-Cyber	1987	4	0.634 [0.61, 0.66]	0.512	0.033	0.774	0.075 [0.06, 0.10]	0.789
LAB-Bench	LB-CloningScenarios	32	4–8	0.469 [0.31, 0.64]	0.314	0.126	0.671	0.500 [0.09, 0.91]	0.579
	LB-ProtocolQA	108	4–7	0.537 [0.44, 0.63]	0.397	0.124	0.620	0.077 [0.01, 0.33]	0.600
	LB-SeqQA	600	2–5	0.618 [0.58, 0.66]	0.492	0.093	0.861	0.000 [0.00, 0.05]	0.825
	LB-DbQA	520	4	0.398 [0.36, 0.44]	0.197	0.086	0.653	0.091 [0.03, 0.28]	0.468
	LB-LitQA2	199	2–10	0.402 [0.34, 0.47]	0.208	0.168	0.656	0.381 [0.21, 0.59]	0.479
	LB-SuppQA	82	4–8	0.366 [0.27, 0.47]	0.195	0.193	0.709	0.250 [0.07, 0.59]	0.429
pooled		6020	2–10	0.673 [0.661, 0.685]	–	0.034	0.820	–	–

WMDP-Bio-Robust (80.1%), and is substantially lower on WMDP-Cyber (63.4%). LAB-Bench results are lower and more variable, from 36.6% on SuppQA to 61.8% on SeqQA; SuppQA is the weakest subtask, though still above its 21.3% chance level. Aggregate accuracy therefore describes the particular task at least as much as the model.

Interpreting the model’s uncertainty requires first establishing what the reported confidence field measures. Across all 6,020 records it matches

$$c = \frac{p_{\max} - 1/n}{1 - 1/n}$$

to within ± 0.015 for 97.9% of records, where n is the number of options. Competing readings fit less well: raw $p_{\max}$ matches 28.0%, the margin between the top two options 31.0%, and normalised entropy 20.7%. The field is a rescaling of the top-1 probability onto $[0, 1]$, mapping a uniform distribution to 0 and a one-hot distribution to 1; on a four-option item an even split between two candidates ($p_{\max} = 0.5$) is reported as 0.33. It is not a predicted probability of correctness, and calibration should be assessed against $p_{\max}$ instead.

On that basis, pooled ECE is 0.034 (95% CI [0.027, 0.045]; Brier 0.161), against 0.091 when the rescaled field is used (Fig. 1(a)). Pooled AUROC for separating correct from incorrect predictions is 0.820 (95% CI [0.810, 0.830]), so $p_{\max}$ carries information about correctness. Both pooled figures flatter the model, since pooling lets $p_{\max}$ discriminate partly on which dataset an item came from: per-task values span 0.62–0.86 AUROC and 0.033–0.193 ECE (Table 1), and those are the relevant figures for a system consuming these predictions. We compared four candidate signals, $p_{\max}$, the top-two margin, negative entropy, and the rescaled field; they differ by at most 0.006 AUROC, so the choice among them is not consequential.

Calibration quality is not uniform. Per-dataset ECE is at most 0.057 on the four WMDP suites but reaches 0.168 on LitQA2 and 0.193 on SuppQA, where item counts are small.

4.2. Confidence and selective prediction

A quantity that is directly interpretable for a system consuming these predictions is the error rate given a confident answer. Under $P(\text{error} \mid p_{\max} \geq 0.9)$ the model is most reliable on WMDP-Bio (3.9%, $n = 848$) and less so on WMDP-Cyber (7.5%, $n = 550$); a Fisher exact test gives $p = 0.00469$, and $q = 0.0141$ after Benjamini–Hochberg correction across all pairwise comparisons. Confident answers are considerably less reliable on the LAB-Bench literature tasks, reaching 38.1% on LitQA2 (95% CI [21, 59]%, $n = 21$), although that subset is small. A confidence threshold calibrated on one dataset should not be assumed to transfer to another.

Withholding low-confidence items raises accuracy on the remainder (Fig. 1(b)). Deferring the least confident 40% of items raises accuracy on the retained set from 85.2% to 97.1% on WMDP-Bio and from 63.4% to 78.9% on WMDP-Cyber. Confidence therefore identifies a subset on which the model performs substantially better, which is a prerequisite for any scheme that would direct difficult items elsewhere.

4.3. Answer-order instability

We next ask whether decisions are stable under a change that does not alter the content of the question. Each item is presented under all four cyclic rotations of its option list, and each returned distribution is mapped back to the original option indices.

Aggregate accuracy is nearly unaffected: across the four rotations it spans 0.8 pp on WMDP-Cyber and 1.3 pp on WMDP-Bio. Item-level agreement is much lower. Only 62.6% of WMDP-Cyber items receive the same answer under all four rotations, against 89.4% on WMDP-Bio (Fig. 2(a)). Error rates differ sharply between the two groups: on WMDP-Cyber, 62.5% of unstable items are answered incorrectly under the original ordering against 21.3% of stable items. A substantial fraction of errors occurs on items whose predictions are unstable under reordering.

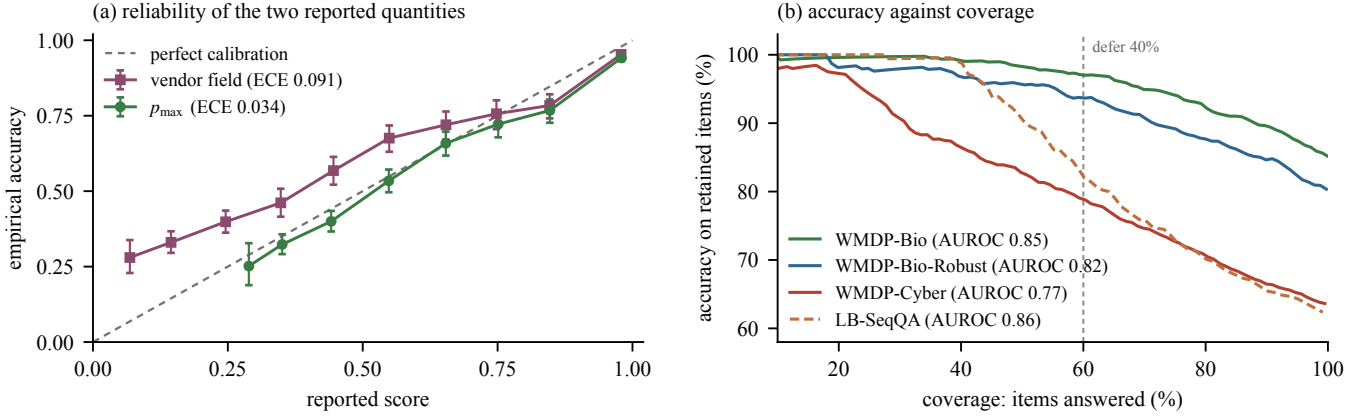


Fig. 1. Uncertainty and selective prediction. (a) Pooled reliability curves for the two quantities the API reports: the confidence field (squares) is an affine rescaling of $p_{\max}$ (circles), so calibration measured against it differs from calibration measured against the probability itself. Bins holding fewer than 25 items are omitted, because $p_{\max} \geq 1/n$ leaves the lowest bins of the $p_{\max}$ curve nearly empty; error bars are Wilson intervals. (b) Accuracy on retained items against coverage, withholding the least confident first, for the three WMDP suites and the largest LAB-Bench subtask (dashed); the remaining six subtasks are omitted for legibility and appear in Table 1. The dashed vertical marker is the 40% deferral operating point quoted in Section 4.2 and in Table 1.

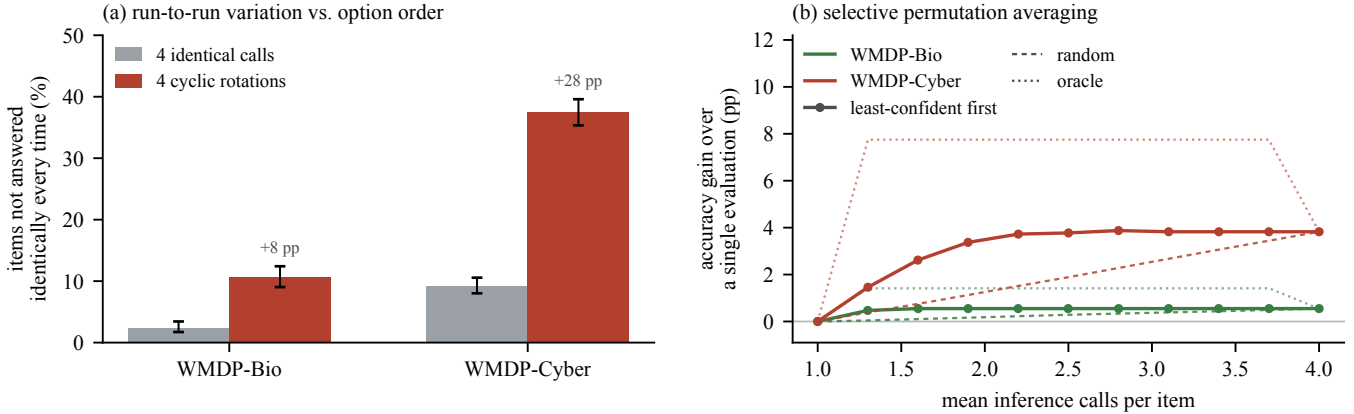


Fig. 2. Answer-order instability and selective permutation averaging. (a) Items answered differently across four byte-identical calls, which isolates run-to-run variation, against four cyclic rotations, which adds option order; the annotation gives the difference. Wilson intervals. (b) Accuracy gain over a single evaluation against mean inference calls per item, plotted as a gain so that two datasets 22 pp apart in absolute accuracy share one scale. Items are selected for additional permutations least-confident first, and compared against random selection at the same budget (averaged over 400 draws) and against an oracle that selects, under a fixed budget, the items where averaging helps.

One observation qualifies this. An association between instability and error is partly implied by the measurement: consistency and correctness are both defined relative to the first rotation, and there is one way to be consistently correct but several ways to be wrong. A null model with no positional preference, matched to the same accuracy and the same consistency, produces an error ratio of 5.43 on WMDP-Cyber (95% CI [4.68, 6.35]) against the 2.93 observed, and 10.03 against 4.64 on WMDP-Bio. The observed ratio falls *below* what the artefact alone predicts. The descriptive result stands; a causal reading does not.

4.4. Separating option order from run-to-run variation

Rotating the options changes the presentation and also makes a fresh call, and the endpoint is not deterministic, so

the second alone could produce disagreement. We therefore held the presentation fixed and queried every item 4 times with a byte-identical prompt, and queried a stratified subsample of 500 items per dataset 3 times at every rotation for a two-factor design (25,040 calls). Dispatch order was shuffled so repeats of one item were never consecutive. Identical prompts disagree far less often than rotations do. On WMDP-Cyber, 9.2% of items receive different answers across 4 identical calls, against 37.4% across four rotations, leaving 28.2 pp attributable to option order; on WMDP-Bio the figures are 2.4% and 10.6% (Fig. 2(a)). Run-to-run variation thus accounts for roughly 25% of the observed instability on WMDP-Cyber and 23% on WMDP-Bio. The two-factor subsample agrees: within-rotation disagreement is 7.8% against 37.8% between rotations on WMDP-Cyber, a ratio of $4.8\times$ ($3.8\times$ on WMDP-Bio). The same design

gives the matched-budget control for Section 4.5: four identical calls cost exactly what four rotations cost. Averaging four identical calls changes WMDP-Cyber accuracy by +0.5 pp, with an interval spanning zero, whereas averaging four rotations gives +3.8 pp; the difference is +3.4 pp (95% CI [+2.0, +4.8]). The gain therefore comes from averaging over presentations, not from averaging away per-call noise. Positional preference is directly detectable too. Since predicted labels are drawn toward correct labels by accuracy, a χ^2 against the correct-label distribution loses power as accuracy rises, so we compare against a simulated null of the same accuracy with no positional preference. The first option is over-selected on 3 of 8 testable datasets — WMDP-Cyber, LB-SeqQA and LB-DbQA — most clearly on WMDP-Cyber ($p < 10^{-4}$; 34.8% of predictions against 26.6% of correct answers) but not on WMDP-Bio ($p = 0.11$; power 1.00 against the worst case in which every error lands on the first option), and on Cyber it is confined to uncertain items: at $p_{\max} < 0.5$, 42.5% of predictions fall first against 21.8% of correct answers ($p < 10^{-4}$), whereas at $p_{\max} \geq 0.9$ it is absent ($p = 0.84$, power 1.00). Because the default prompt labels options in positional order, a preference for the first *position* and one for the letter *A* are confounded; two re-runs separate them. Replacing the letters with non-alphabetic glyphs, which carry no ordinal prior, leaves the effect intact, the first position taking 36.1% of WMDP-Cyber predictions against 34.8% at baseline. Permuting the letters across positions splits it, with 30.5% falling on the first position and 31.2% on the letter A. The preference is therefore primarily positional, with a weaker label component. Quantisation ties do not explain it either: ties occur on only 1.2% of records (71 items, the first option among the tied set in 46), and it wins 21 of them against 23.0 expected under random tie-breaking. Accuracy is barely affected by the labelling scheme (glyphs change it by -1.0 pp, 95% CI $[-2.4, +0.5]$), but prompt format matters: listing the options only as category keys costs 4.7 pp on WMDP-Cyber (95% CI $[-6.3, -3.1]$) and listing them only in the state text costs 2.0 pp ($[-3.3, -0.8]$), so the redundant serialisation is doing work.

4.5. Selective permutation averaging

If a single evaluation returns one of several possible answers, averaging over presentations should help where instability is high. Averaging the four inverse-permuted probability distributions raises WMDP-Cyber accuracy from 63.3% (rotation 0 of the cyclic run; the independent single-pass figure in Table 1 is 0.634) to 67.1% (+3.8 pp, 95% CI [+2.4, +5.3]); majority voting over the four answers is weaker (65.2%). On WMDP-Bio, where 89.4% of items are already stable, the gain is +0.5 pp with an interval spanning zero ($[-0.2, +1.4]$). Averaging helps on the dataset that exhibits substantial instability and not on the one that does not.

Evaluating four rotations costs four times as much as one. Since $p_{\max}$ predicts correctness, it can be used to select which items receive the additional evaluations. Presenting

a fraction c of items under all four rotations, chosen least-confident first, costs $1+3c$ calls per item. On WMDP-Cyber, selecting the least confident 20% reaches 65.9% and the least confident 40% reaches 67.0% at 2.2 calls per item, which is 97% of the gain available from averaging every item at 45% lower cost (Fig. 2(b)). Random selection at the same budget reaches 64.8%, averaged over 400 draws, and no draw beat the confidence-based ordering. On WMDP-Bio the advantage over random selection is +0.3 pp and 6.8% of random draws do at least as well, so we do not claim a benefit there. The distance to the oracle curve indicates the headroom remaining for a better selection rule.

5. Discussion and Limitations

These results bear on how System-1 models should be evaluated. Aggregate accuracy was nearly invariant across the four rotations while more than a third of WMDP-Cyber items received different answers, so similar accuracy figures can conceal substantial item-level instability. Reported uncertainty is informative once its definition is established: $p_{\max}$ separates correct from incorrect predictions with pooled AUROC 0.820 and supports abstention, but calibration quality and the error rate at high confidence both vary by dataset, so a threshold chosen on one task should not be assumed to hold on another. Stability under re-presentation is a second signal, unavailable from a single evaluation, and repeated inference is worth its cost where instability is high but not where it is low.

Where users can influence how a decision is presented, representation sensitivity could become a manipulation surface, and disagreement across re-presentations could signal an unreliable decision. Our benchmark does not test such a deployment.

These properties are relevant if System-1 models are later considered as components of biosecurity classifiers or language-model safety pipelines. Evaluating such a use would require a dedicated benign-versus-hazardous classification benchmark, which this study does not provide.

5.1. Limitations

We evaluate one proprietary closed-weight model that exposes no seed, so these results do not generalise to System-1 models as a class and exact reproduction depends on vendor behaviour. We evaluate no biosafety or hazard-classification task, and multiple-choice questions are only a proxy for the knowledge they probe; label quality in biology-adjacent sets can be poor [19] and we did not audit it, nor can benchmark contamination be excluded. We did not measure latency and did not run an external larger model as a fallback, so we make no claim about either. Cyclic evaluation covers four rotations rather than all 4! permutations and only the four-option WMDP suites, making the reported instability a lower bound. The label and format ablations were run on those two suites only, so their conclusions do not automatically extend to the LAB-Bench subtasks. Returned probabilities are quantised

to two decimals, and several LAB-Bench high-confidence subsets contain fewer than 25 items, so we report intervals throughout.

6. Conclusion

We audited a commercial non-generative System-1 model on 6,020 items drawn from biosecurity-relevant and biology-relevant benchmarks. Its accuracy is strongly task-dependent, its top-1 probability is reasonably well calibrated and predicts its own errors well enough to support abstention, and its decisions are considerably less stable under reordering of the answer options than aggregate accuracy suggests. Averaging probabilities over answer permutations improves accuracy where instability is high, and applying it only to low-confidence items recovers most of that improvement at a fraction of the cost. Confidence and stability under re-presentation provide complementary evidence about when such a model’s predictions should be trusted, and neither is visible in a single accuracy figure.

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Ethics and Conflict of Interest

This study authored no new hazardous technical content: all items come from existing public benchmarks, and we release model decision records but no new question content. We reported the `confidence` field semantics to the vendor as a documentation issue. The authors declare no competing financial or non-financial interests.

Code Availability

Decision records, analysis code, and manuscript source are available at https://github.com/Georgakopoulos-Soares-lab/biosafety_knowledge_jev. Every reported statistic can be recomputed from them.

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LLM Usage Statement

A large language model (LLM) based coding assistant (Anthropic Claude) was used to write the evaluation harness and analysis scripts, to assist with statistical analysis and literature search, and for editorial work on this manuscript.

All generated code was executed and its outputs inspected by the authors. Every statistic in the running text is emitted by the released pipeline, and the build fails if a cited value does not match it; all references were verified against primary sources. An earlier version of this analysis contained statistical errors, found and corrected during internal adversarial review. The model under evaluation is a commercial closed-weight system, a reproducibility limitation we cannot remove. The authors take full responsibility for the correctness, originality, and integrity of all content.